

Achieving Water Efficiency Through Performance Contracting

by Tom Horner

As the vice president of a company that helps building owners and operators all across the country save water, I'm confronted with the magnitude of our thirst for water and the costs to meet that thirst every day. The statistics on the nation's water use are quite eye opening! Right now withdrawals for drinking water from public water sources are approximately 45 billion gallons per day—that's up from 15 billion gallons per day in the 1950s. And slightly more than 85% of the U.S. population, or about 255 million people, get their drinking water from public suppliers. That number continues trending upward. Between 1995 and 2000, public-supply withdrawals and the population served increased by 8%. States with the largest populations withdraw the largest quantities of water; California, New York, Michigan, Florida, North Carolina, and Illinois are the water use leaders.

Our water and sewer delivery system is one of the best in the world, and this has surely helped to make our country great, and has helped to make our economy grow and flourish. Life as we know it in the western United States would be impossible today without the enormous ingenuity and effort it takes to supply clean water to cities such as Las Vegas, Phoenix, and Los Angeles. But attaining that level of service requires large amounts of energy. Water and wastewater treatment facilities are some of the largest and most energy-intensive properties owned by local governments. The energy costs to pump, treat, and deliver water, and then to collect and treat wastewater, represent at least 25% of a typical municipality's energy bill. At an average energy cost



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of approximately \$0.35 per thousand gallons, our country's energy bill for pumping water and wastewater is more than \$5 billion per year.

With energy prices rising, the growing need to improve and repair water delivery and sewage infrastructure, and the demand for water constantly increasing, it's not hard to figure out that significant rate increases are right around the corner. Providing clean, adequate water supplies and safe, efficient sewage treatment while controlling rising costs is a tremendous challenge for our nation's approximately 54,000 water delivery and treatment systems.

MEETING DEMAND WHILE NOT GOING BROKE

Like politics, all water is local; there are few interconnected systems. And systems don't exist that allow one to buy water off the grid the way utilities buy electricity wholesale.

Water just may be one of the last great monopolies in America. Water authorities in growing communities often need a way to help offset the consumption anticipated by growth, because their existing systems and infrastructure are at or near capacity. It is usually cheaper to save water than it is to produce more. Using water more efficiently allows water and sewer authorities to avoid the upgrades needed for extra capacity, and save the construction costs involved. And since the great majority of the nation's 54,000 water authorities are owned by the towns and cities they serve, programs that save water are really saving us our own money. Building owners, residents, and local sewer authorities all benefit from water and energy efficiency.

Water efficiency has become one of the most successful tools that water and sewer providers are using to combat ever-increasing demand. Education programs, coupled with rebates, are a more cost-effective method of meeting demand than much costlier new-plant construction. New York City's Toilet Rebate program, which began in 1994, is a prime example of a water-conserving program that works. This initiative has installed more than one million toilets and cut water use in the city by about 60 million gallons a day. These savings have allowed New York to avoid the much higher costs of building new water supply and wastewater treatment capacity.

STEPS TO TAKE

Residential water use generally consists of kitchen and bathroom uses combined with laundry and maybe some cooling or irrigation. When contemplating water efficiency

retrofits, the first step that owners or property managers need to take is to quantify the true water usage in their buildings and the real costs involved for maintenance and operations. Then realistic baselines can be arrived at and individual water conservation measures can be analyzed for savings and practicality.

The individual conservation measures for residential properties would generally fall in the following categories:

- Upgrade all dishwashers and clothes washers to water-efficient appliances.
- Replace components of older plumbing systems with long-term leak-free water-saving equipment such as ultralow-flow toilets, high-efficiency showerheads, and aerators.
- Ensure that all tub and shower assemblies use pressure balance valves to use the minimum amount of water needed to get the desired temperatures.
- Maintain proper pressure at the property through the use of pressure-regulating valves to ensure that the flow out of end use devices is only as much as is needed to perform the desired task.
- Replace once-through cooling devices with closed-loop or air-cooled systems.
- Audit steam systems for trap leakage and recirculation problems.
- Upgrade irrigation system components and control usage with integrated evapotranspiration sensors and systems.
- Install graywater, rainwater, groundwater, and reclaimed water recycling technology for toilet flushing, irrigation, or cooling tower use. Throughout the state of Florida, counties use graywater distribution systems for irrigation;

this significantly reduces the demand for potable water, and the water is delivered and sold to the end user at a significantly reduced fee.

- Install monitoring equipment and submeters as needed so that increases in consumption over time can be quickly rectified.

Building owners have an excellent opportunity to reduce operating expenses and lower flows at treatment facilities while helping to conserve a natural resource.

PERFORMANCE CONTRACTING

As with other major energy efficiency upgrades, the method most often recommended to private-sector building owners who wish to fund a total water management program is performance contracting. Performance contracting is a method of minimizing risk through guaranteed savings. When used in the water efficiency field, it covers a range of agreements where the original and ongoing costs of reducing water and sewer expenses in a facility or multifamily residence are paid for out of savings generated by the program.

Performance contracting has been used in the government and in the private sector. The vast majority of programs in the government arena are run by energy services companies (ESCOs). These companies are well versed in program development and in bundling together all efficiency conservation measures under one program. They have been very successful in the federal market, in state and local governments, and in educational systems. Water efficiency is one of the many opportunities that are included in these comprehensive energy programs.

The most utilized contracts for the federal market are super energy savings performance contracts (ESPCs) and utility energy service contracts. These contracts are preapproved by the government, and all the processes and procedures are defined. The utility

energy service contract program began in 1995, and the super ESPCs program began in 1998. The Energy Policy Act of 2005 reauthorized all performance contracting through 2016.

More than 400 ESPC projects have been awarded by 19 different federal agencies in 46 states, and \$1.9 billion has been invested in U.S. federal facilities through these contracts. More than 45 electric and gas utilities have provided project financing for energy and water efficiency upgrades at federal facilities, investing more than \$600 million through utility energy service contracts since 1995.

In the private sector, apartment owners, real estate investment trusts (REITs), and commercial property owners have realized the relatively short paybacks of water efficiency programs and have upgraded their plumbing in most high-water-rate areas to low-flow equipment. After several successful projects, the stakeholders realized that there were low-flow plumbing fixtures (toilets, faucets, and showerheads) available that exceeded the performance of models that used more water. They realized that with the proper product selection, toilets would not clog and actually flushed better than the current 3.5- to 5-gallon units in their buildings. With a minimum of 20% in savings, most facilities showed return on investments in less than five years.

Major plumbing equipment upgrade projects have been funded by a variety of techniques, from direct payment to long-term loans. The two main types of performance contract that are currently in use are shared savings and guaranteed savings.

In shared-savings agreements, the water service company pays all program and long-term costs, including the cost of products, installation, monitoring, and maintenance. Their payment is based on a portion of the savings achieved, thus minimizing the risk to the owner. This is usually an agreed-upon percentage of the direct cost savings from the agreed-upon baseline figures. In essence, the facility is paying a

reduced fee for nega-gallons (virtual water saved by the program).

In a guaranteed savings agreement, the water service company implements the program for an agreed-upon up-front fee. The company then guarantees that a certain level of savings will be achieved for the duration of the contract—typically for 5 to 20 years.

Although performance contracts for reducing water and sewer costs at a facility are typically less expensive than contracts for energy savings, the cost savings generated can be as high if not higher. Factors that affect the level of savings are the property's current consumption level and its water and sewer rates. Throughout the United States, these rates vary dramatically, depending on how the water and sewer infrastructure was funded. With recent changes in how water and sewage must be treated before they are released into rivers, streams, or other water sources, some local rates have increased over 200% over the last five years. Water rates in Atlanta, Boston, and Washington, D.C., for example, are 2 or 3 times the national average per 1,000 gallons. Although water is still one of life's greatest bargains at a delivered price of less than a penny a gallon (in over 95% of municipalities), the savings potential in implementing a well-designed water efficiency program is enormous.

SPENDING MONEY TO SAVE MONEY

The simple solutions for cutting water and sewer costs in commercial and multifamily buildings were instituted in the '90s and in the early years of this century. More comprehensive solutions can now be funded as water and sewer rates have gone up. The most successful option involves a whole-building or main-meter approach. It's possible to guesstimate or engineer what individual components of a program will save, but most property owners are only interested in the bottom line. A long-term program cannot be funded out of even reliable estimates; it must

be funded out of true accounting figures.

All aspects of water usage must be combined into the program. Someone must take total responsibility for monitoring actual water and sewer consumption and savings, as measured by the water and sewer authority, if performance contracting is to be successful.

The following formula can be used to compute cost savings:

$$\text{Water and sewer cost savings} = \text{Baseline water and sewer usage} \times \text{Present rate} - \text{Postinstallation water and sewer usage} \times \text{Rate.}$$

To establish a figure on which to base all savings calculations, the agreed-upon method is to average the previous one to three years of consumption (directly from past water and sewer bills) and to convert this number into units per day of usage. These units will typically be gallons, but they can also be cubic feet. The baseline figure will be expressed in units of water and sewer use, not in dollars. During the term of the performance contract, this baseline figure will be converted into a dollar amount by using the current water and sewer rate in that jurisdiction.

One main advantage of this approach is that the quality of the products installed can be significantly better than it is when the property owner is looking for the quickest return on investment. Ceramic-disk faucets; pressure balance shower valves; and high-quality toilets, showerheads, and aerators can be installed that function better than the original products they replace. Pressure-regulating valves and submeters could be added to further control and monitor consumption levels. Engineering analysis of each site could be performed to ascertain where new technologies and management techniques for cooling towers, boilers, laundries, ice making, dish washing, and special purposes can be implemented. Landscaping water use can be greatly reduced by using xeriscaping and more efficient irrigation practices.

The findings from years of water efficiency programs are conclusive. Savings of approximately 20% can be achieved temporarily with almost any off-the-shelf toilet combined with leak repair and standard showerheads and aerators. But for the program to be perceived as a significant upgrade for the users, the quality of the products must be better than the norm. Spending more for quality on the front end will achieve long-term savings of at least 30%, with maintenance costs reduced by well over 50%.

The plumbing industry is similar to many others. The quality of the products available runs the full range. Over the 25 years that our company has been in business, we have found many 5-gallon, 3.5-gallon, and 1.6 gallon-per-flush toilets that performed horribly. We have also found some 1-gallon and 1.6-gallon toilets with superior flushing dynamics and high-quality parts and fittings.

The energy costs to deliver and treat water represent a portion of your water bill. Water authorities need to upgrade their pumps to variable-speed drive models and track their systems to eliminate inefficiencies. At an average energy cost of approximately \$0.40 per 1,000 gallons, our country's energy bill for pumping water and waste water is over \$6 billion per year. Using our water more efficiently could cut our overall use by 20%. Do you think it would be impossible to reduce use from 150 gallons to 120 gallons per person per day? This represents a savings of 15 billion kilowatt-hours of electricity, and a corresponding reduction in green house gas emissions.

Remember the big picture—safe, adequate water and sewer treatment services are a big part of what makes our country great. Saving our natural resources through well-designed programs benefits our whole country.



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